machine learning

stats Algorithms

Data → model

Regression

classification

Algo – 100

clustering

feature

	x_1	x_2	x_3	x_4	Target
1	–	–	–	–	A
2	–	–	–	–	B
3	–	–	–	–	C
					D

– – – – B

/A/C/D

Data

Lable data

unlable data

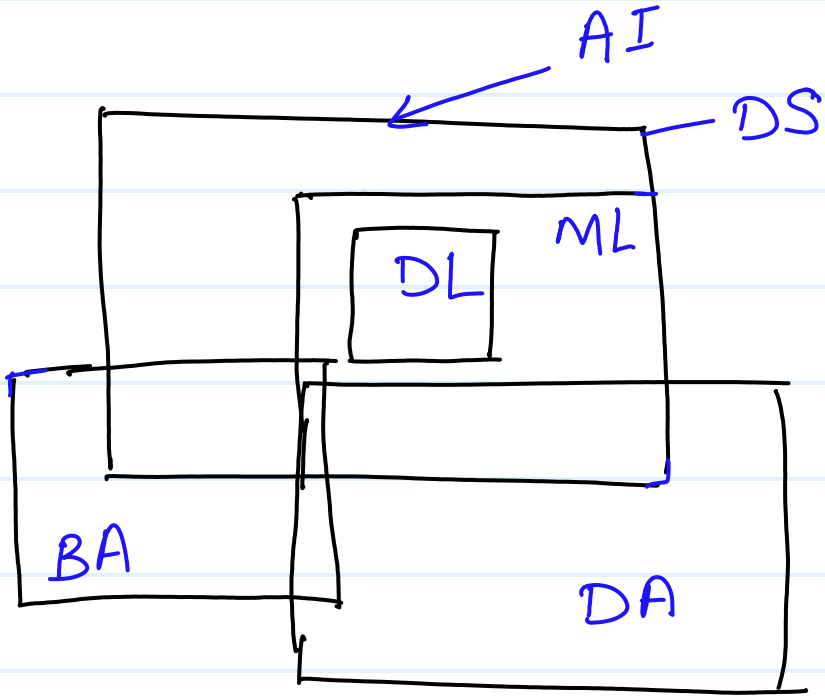
feature Target

X Y

Ind col dep. col.

X Y

X_1 X_2 X_3 X_4



Machine Learning

Supervised ML

↓

Lableddata

Unsupervised ML

↓

Unlable data

Supervised ML

Regression | classification

$X_1 \; X_2 \; X_3$ $\quad$ Y ← cont. variable

Regression

categorical

classification

Regression		classification
I) Linear Reg. / L_1 / L_2 / EN		Logistic Reg.
II) SVR ←	SVM	→ SVC
III) KNNR ←	KNN	→ KNNC
IV) Polynomial Reg.		Naiv bay's
V) DTR ←	DT	→ DTC
VI) RFR ←	RF	→ RFC
VII) Ada boost R ←	AB	→ Adaboost C
VIII) Gradient Boosting —	GB	— Gradient boosting
IX) XGBoost —	XGB	— XGBoost

⊗ Catboost — CB — Catboost

Unsupervised ML

Clustering / Grouping

I. k-mean clustering.
II. k-mean ++.
III. Hirerchical clustering.
IV. DBScan.
V PCA.
VI. Apriory Algorithms

Time Series

Forecasting [Timeseries data]

I. ARIMA.
II SARIMA.
III. SARIMAX

X Y

X, X, .. Y.

ARIMA (P, d, q)

→ (1, 1, 1)

→ (1, 1, 0)

→ (0, 1, 1)

→ (2, 1, 0)

→ (2, 2, 1)

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I. - Data

II. - Outlier, Noise, Anomalies

III. - Feature Engg.

IV. - Feature selection

V. - Feature encoding

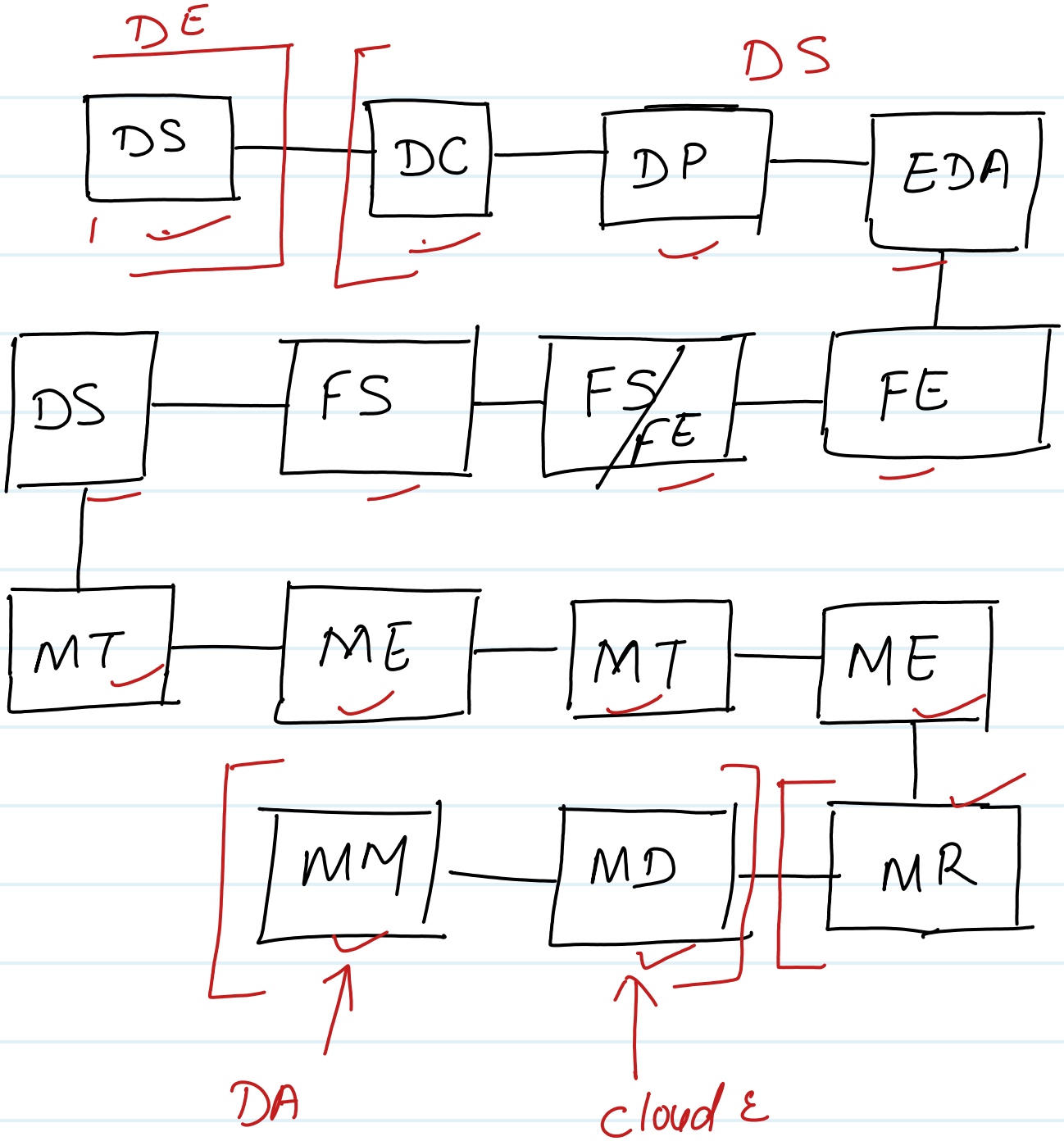
VI. - Feature scaling

VII. - Data split (data leakage)

VIII - Regularization / Generalization

IX. - Model Tuning / Hyperparameter Tuning

X - Data drift



'DL'

① Reinforcment learning → Train
→ Retraining

② semi supervised